Ayesha Mashiat

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Education

- **Islamic University of Technology** — Expected 2027
  B.Sc. Software Engineering — *Gazipur, Bangladesh*
- **South Point College** — 2022
  HSC – GPA 5.00/5.00 — *Dhaka, Bangladesh*
- **South Point School** — 2020
  SSC – GPA 5.00/5.00 — *Dhaka, Bangladesh*

Experience

- **Software Engineering Intern** — Sep 2026 – Present
  RedDot Digital Limited — *Dhaka, Bangladesh*
  - Working as an intern under the AI and Automation Team and contributing to Smart ECRM project of Smart Cambodia.
  - Gaining hands-on experience in software engineering and real-world business applications while working under the guidance of a Lead Software Engineer.
- **Frontend Developer (Volunteer)** — 2026
  Project Beaver (STEM Muslims) — *Remote*
  - Contributed React.js components to an open-source room availability platform integrating Imperial College London's CELCAT system.
  - Built the room search interface with date, time, department, and room filters while collaborating with an international volunteer team using Git and Figma.
- **Backend Developer (Volunteer)** — 2026
  Sahbti (Tech Sisters Hackathon 2024 Winning Project) — *Remote*
  - Developed authentication functionality for an AI-powered application using FastAPI and MongoDB, contributing to the backend of a collaborative open-source project.
  - Implemented user registration, login, and JWT-based authentication while coordinating backend changes through Git and GitHub.

Projects

- **Jobscape – Job Portal Platform** | *FastAPI, PostgreSQL, SQLAlchemy, Alembic, spaCy, Groq API, Next.js, JWT, Cloudinary* | GitHub

  - Built a full-stack job platform with a FastAPI backend and Next.js frontend supporting resume parsing, job matching, and real-time video interviews.
  - Implemented Google OAuth2 and JWT authentication with refresh tokens, email verification, and API rate limiting using SlowAPI.
  - Designed the backend data layer with SQLAlchemy and Alembic migrations, and added APScheduler background jobs and Cloudinary media storage.

- **CareerPilot – AI Career Co-Pilot** | *FastAPI*, *Groq API*, *ChromaDB*, *Sentence-Transformers*, *Redis*, *MongoDB*, *React*, *Docker*, *Nginx* | GitHub
  - Built a RAG-based career assistant using Sentence-Transformers for embeddings, ChromaDB for vector retrieval, and Groq API for context-aware responses.
  - Developed an AI CV generator that adapts resumes to job descriptions and a voice interview coach providing real-time interview feedback.
  - Built a containerized FastAPI backend with MongoDB persistence, Redis session management, and Nginx reverse proxy configuration.

- **Touch Grass, Babu – Offline AI Digital Wellbeing App** | *Python*, *Qwen3.5-4B*, *LoRA*, *GGUF*, *llama.cpp*, *Android NDK/JNI*, *ML Kit* | GitHub
  - Fine-tuned a 4B language model using LoRA and a curated synthetic dataset, improving rule-following from 33% to 93% across 30 held-out evaluation scenarios while reducing prompt usage from 3,675 to 41 tokens.
  - Quantized the model to GGUF Q4_K_M and integrated local inference into Android through llama.cpp and JNI, achieving approximately 1.4 tokens/sec on budget hardware through CPU-thread optimization.
  - Built an offline digital wellbeing system combining app-usage monitoring, background roast pre-generation, step counting, ambient-light sensing, and on-device image labeling to enforce screen-time restrictions.

Technical Skills

- **Languages:** Python, JavaScript, TypeScript, Java, C, C++, SQL
- **Backend:** FastAPI, Node.js, Express.js, REST APIs, SQLAlchemy, JWT, OAuth2
- **Databases:** PostgreSQL, MongoDB, MySQL, Oracle
- **AI/ML:** RAG, Sentence-Transformers, ChromaDB, spaCy, LLM APIs
- **Infrastructure:** Docker, Nginx, Redis, AWS
- **Tools:** Git, GitHub, Postman, Alembic, APScheduler, Cloudinary

Awards & Honors

- **Finalist – IUT 12th ICT Fest** 2026
  Ranked among the Top 20 teams out of 700+ participating teams.
- **Finalist – CodeSprint Inter-University Hackathon** 2026
  Ranked among the Top 14 teams out of 50 participating teams.
- **Finalist – SUST CSE Carnival Codex Community Hackathon** 2026
  Ranked among the Top 50 teams out of 450+ participating teams.
- **Solvio AI Hackathon 2025 – Phase 2 Qualifier** 2025
  Advanced to Phase 2 after ranking among the Top 100 teams.